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Prob & Stats Assignment 2

Q2) Binomial random variable - $B(n,p)$

Mode is the value of X for which $P_p(n)$ (PMF) is maximum.

Let $a_k = P(X=k)$ we have

$a_k = \binom{n}{k} p^k (1-p)^{n-k}$ and $a_{k+1} = \binom{n}{k+1} p^{k+1} q^{n-k-1}$

~~where as usual q = 1~~

let $q = 1-p$ for a binary dist.

So, $\frac{a_{k+1}}{a_k} = \frac{\binom{n}{k+1}}{\binom{n}{k}} \frac{p^{k+1} \cancel{q^{n-k-1}}}{\cancel{p^k q^{n-k}}} = \frac{\binom{n}{k+1}}{\binom{n}{k}} \frac{p}{q}$

So, $\frac{a_{k+1}}{a_k} = \frac{n-k}{k+1} \cdot \left(\frac{p}{q}\right) = \frac{n-k}{k+1} \left(\frac{p}{1-p}\right)$

We can follow,

$k > (n+1)p - 1 \Rightarrow a_{k+1} < a_k$

$k = (n+1)p - 1 \Rightarrow a_{k+1} = a_k$

$k < (n+1)p - 1 \Rightarrow a_{k+1} > a_k$

So, when $(n+1)(p) - 1$ is an integer then modes are $[(n+1)p] - 1$ and $[(n+1)p]$ but, when $(n+1)p - 1$ is not an integer then mode is $[(n+1)p]$.

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Q3) $Var(ax+b) = E[(ax+b)^2] - (E[ax+b])^2$

$\Rightarrow E[a^2x^2+b^2+2axb] - a^2(E(x))^2 - b^2 - 2abE[x]$

$= a^2E[x]^2 + \cancel{b^2} + \cancel{2ab\,E[x]} - a^2E(x)^2 - \cancel{b^2} - \cancel{2abE[x]}$

$Var(ax+b) = a^2[E[x^2] - E[x]^2] = a^2\,Var(x)$

$\therefore$ Hence, proved

Q4) $P(X=n) = {}^{n-1}C_{r-1}(p)^r(q)^{n-r}$

not accounted for the rth head since it would have a fixed position. The last flip.

H T H . . . . H | game ends.
r^{th}

Ans $= {}^{n-1}C_{r-1}\,p^r(1-p)^{n-r}$

Q5) i) $P[X=k \mid X+Y=n] = \dfrac{P(X=k \cap X+Y=n)}{P(X+Y=n)}$

$$\frac{P(X=k \cap Y=n-k)}{P(X+Y=n)} = \frac{\dfrac{\lambda_1^k e^{-\lambda_1}}{k!} \times \dfrac{\lambda_2^{n-k} e^{-\lambda_2}}{(n-k)!}}{P(X+Y=n)} \quad \text{— (1)}$$

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Now, $P(X+Y=n) = \sum_{k=0}^{n} \frac{\lambda_1^k e^{-\lambda_1}}{k!} \cdot \frac{\lambda_2^{n-k} e^{-\lambda_2}}{(n-k)!} \cdot {}^nC_k$

$= e^{-\lambda_1} \cdot e^{-\lambda_2} \sum_{k=0}^{n} \frac{{}^nC_k \times \lambda_1^k \times \lambda_2^{n-k}}{k!\,(n-k)!}$

$= e^{-\lambda_1-\lambda_2} \times (\lambda_1+\lambda_2)^n$

So, $P(X=k \mid X+Y=n) = \frac{\lambda_1^k \lambda_2^{n-k}}{k!\,(n-k)!} \times e^{-\lambda_1}$

$= \frac{\lambda_1^k \lambda_2^{n-k}}{k!\,(n-k)!\,(\lambda_1+\lambda_2)^n}$

(ii) $E[X \mid X+Y=n] = \sum_{k=0}^{n} \frac{k \times \lambda_1^k \times \lambda_2^{n-k}}{k!\,(n-k)!\,(\lambda_1+\lambda_2)^n}$

$= \frac{\lambda_1}{(\lambda_1+\lambda_2)^n} \sum_{k=1}^{n} \frac{\lambda_1^{k-1} \times \lambda_2^{(n-1)-(k-1)}}{(k-1)!\,(n-1-(k-1))!}$

$= \frac{\lambda_1}{(\lambda_2+\lambda_1)^n} \times (\lambda_1+\lambda_2)^{n-1} = \left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right) E[X \mid X+Y=n]$

Q6) $X \sim Exp(\lambda)$. Find $E[e^{tx}]$, $t < \lambda$

pdf $= \lambda e^{-\lambda x}$

$E[e^{tx}] = \int_0^\infty e^{tx} \lambda e^{-\lambda x}$

$= \int_0^\infty e^{(t-\lambda)x} \lambda \, dx$

$= \frac{\lambda \left[e^{(t-\lambda)x}\right]_0^\infty}{t-\lambda} = \frac{\lambda}{t-\lambda}\left[0-1\right] = \boxed{\frac{\lambda}{\lambda-t}}$

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Q7) $X_1 \sim N(\mu_1, \sigma_1^2)$
$X_2 \sim N(\mu_2, \sigma_2^2)$

$E[e^{t(X_1+X_2)}] = E[e^{tx_1} \times e^{tx_2}]$

$\therefore$ Since, they are independent

$\int e^{tx_1} \cdot e^{\frac{(x_1-\mu)^2}{2\sigma^2}} \times \int e^{tx_2} e^{\frac{(x_2-\mu)^2}{2\sigma^2}}$

$= e^{\mu_1 t} \times e^{\frac{1}{2}\sigma_1^2 t^2} \times e^{\mu_2 t} \times e^{\frac{1}{2}\sigma_2^2 t^2}$

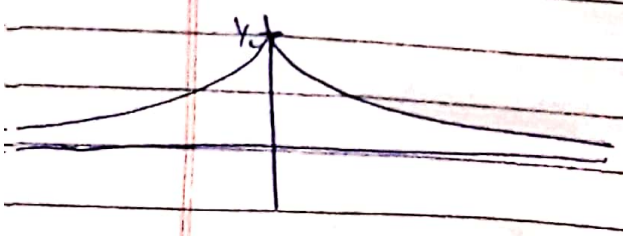
$= e^{(\mu_1+\mu_2)t} \times e^{\frac{1}{2}t^2(\sigma_1^2+\sigma_2^2)}$

$= \left[e^{(\mu_1+\mu_2)t + \frac{t^2}{2}(\sigma_1^2+\sigma_2^2)}\right]$

Q1) $f(x) = \frac{1}{2} e^{-|x|}$

(i) $cdf = \begin{cases} \int \frac{1}{2} e^{x} & x < 0 \\ \int \frac{1}{2} e^{-x} & x > 0 \end{cases}$

$= \begin{cases} \left[\frac{1}{2} e^{x}\right]_{-\infty}^{0} = \frac{1}{2}\left[e^{x} - e^{-\infty}\right] = \frac{1}{2} e^{x} & x < 0 \\ 1 - \frac{1}{2} e^{-x} & x > 0 \end{cases}$



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(iii) (a) $\{|R| \le 2\}$ $\therefore$ $-2 \le R \le 2$

$$\int_{-2}^{0} \tfrac{1}{2} e^{x} + \int_{0}^{2} \tfrac{1}{2} e^{-x}$$

$$= \tfrac{1}{2}\left[1 - e^{-2}\right] + \tfrac{1}{2}\left[e^{-2} - 1\right]$$

$$= 1 - \tfrac{1}{2} e^{-2} - \tfrac{1}{2} e^{-2} = \boxed{1 - e^{-2}}$$

(b) $2R - 3 \le 3$ when $R \geq 3$

$R - R \le 3$ and $0 \le R \le 3$

$3 - 2R \le 3$ and $R < 0$

$P(0 \le R < 3) = 1 - \tfrac{1}{2} e^{-3} - \tfrac{1}{2} = \tfrac{1}{2} - \tfrac{1}{2} e^{-3}$

(c) $\{R^3 - R^2 - R - 2 \le 0\}$

$(R^2 + R + 1)(R - 2) \le 0$

$2R + 1 = 0$

$R = -\tfrac{1}{2}$

$R \geq 2$

always negative

no real roots.

Ans $1 - \left[1 - \tfrac{1}{2} e^{-2}\right] = \tfrac{1}{2} e^{-2}$

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